

Zixu Wang

MSc Artificial Intelligence • Lancaster University

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Profile

MSc AI student with applied work across explainable medical AI, multi-agent LLM simulations, machine learning, NLP and data-driven software systems. Research interests span multimodal vision–language models and LLM-based agents (tool-use, planning and reasoning). Strong focus on turning model outputs into auditable, quantitative evidence and building reproducible analysis pipelines.

Education

Lancaster University

MSc Artificial Intelligence

Lancaster, UK

Sep 2025 – Sep 2026

Average-to-date: 79.4% (Distinction, indicative). Modules: Machine Learning, NLP and Language Models, Intelligent Agents and Autonomous Systems, Advanced Topics in AI, Data Science and AI Fundamentals.

Aston University

BSc Computer Science (Hons)

Birmingham, UK

Sep 2021 – Jul 2024

Second Class Honours. Selected modules: Mathematics (79.0), Human–Computer Interaction (70.0), Team Project, Internet Applications & Database Design, Image & Video Processing.

Research

MSc Dissertation • Toward Explainable Face2Gene Diagnosis

Jun 2026 – Sep 2026

Lancaster University, supervised by Dr. Richard Jiang | NumPy • XAI • Medical AI

- **Built:** a reproducible coarse-ROI Anatomical Evidence Vector (AEV) pipeline for explainable facial phenotyping, converting occlusion-based model behaviour into quantitative ROI-level evidence on low-resolution 32×32 facial images.
- **Engineered:** a dependency-light NumPy MLP workflow on Face2Brain / PD-DBS face data covering QC, calibration, ROI masking, FDR-corrected statistics, leakage/confounding audits and robustness checks.

Neglected Tropical Disease Clinical Trials Analysis

Sep 2025 – Dec 2025

SCC.450 Introduction to AI and Data Science | Python • Regression • Network Analysis

- **Led:** data cleaning for 315 WHO ICTRP trial records across 62 countries; built logistic-regression models, sponsor–country–disease network analysis and publication-quality visualisations.

Projects

Image Classification & Clustering

PyTorch • DINOv2 • Clustering

SCC.451 Machine Learning

- Applied transfer learning and fine-tuning with ResNet50, DenseNet121 and DINOv2 for image classification; used K-means, hierarchical clustering and PCA for climate-data analysis.

Diversity Beats Strength? Multi-LLM Teams in Game-Theoretic Settings

LLM Agents • Simulation

SCC.452 Intelligent Agents

- Implemented and configured a modular multi-LLM evaluation architecture, testing GPT-4o, DeepSeek-V3 and Gemini-3-Flash across Prisoner's Dilemma, Snowdrift and Stag Hunt.
- Ran and organised approx. 6,400 trial simulations, analysing cooperation, payoff, cheap-talk, memory effects, network topology and team composition.

Sentiment Analysis across Multilingual Datasets

NLP • English/Chinese

SCC.453 NLP and Language Models

- Compared tokenisation, stemming and lemmatisation strategies for English and Chinese sentiment-classification pipelines, evaluating preprocessing impact on accuracy.

Undergraduate Projects: Restaurant Back-Office System & PC Components E-Commerce Site

Aston University

Individual / Team

- Built a restaurant backend/admin management system (final-year individual project) and a team-based e-commerce site for PC components, contributing to requirements, implementation, testing and delivery.

Technical Skills

- **Programming:** Python, SQL, Java
- **AI / ML:** PyTorch, scikit-learn, NumPy MLPs, transfer learning, clustering, classification, model evaluation
- **NLP / LLMs:** Text preprocessing, OpenAI / Anthropic / Google API integration, agent simulations
- **Data / XAI:** pandas, NumPy, matplotlib, seaborn, NetworkX, occlusion analysis, calibration, AUROC/AUPRC
- **Tools & Languages:** Git, Linux, Jupyter, LaTeX; Chinese (native), English (fluent)